

Transaction Fraud Analytics and Risk Analysis

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Abstract

This project presents a Transaction Fraud Analytics and Risk Analysis System developed using MySQL and Power BI. The objective is to analyze transaction patterns, identify fraud-prone behaviors, and provide actionable business insights through interactive dashboards. The system evaluates fraud trends across merchant categories, countries, transaction channels, authentication methods, and user risk profiles. By transforming raw transaction data into meaningful visualizations, the solution enables organizations to monitor fraud activity, identify high-risk users, and improve fraud prevention strategies.

1. Introduction:

Financial fraud continues to be one of the major challenges faced by digital payment platforms and financial institutions. Fraudulent transactions result in financial losses, reduced customer trust, and increased operational costs.

The goal of this project is to develop a fraud analytics platform capable of:

- Monitoring transaction activity
- Measuring fraud occurrence rates
- Identifying high-risk users
- Analyzing authentication effectiveness
- Detecting fraud patterns across regions and merchant categories
- Supporting business decision-making through dashboard

2. Objectives:

The main objectives of this project are:

1. Analyze transaction data to identify fraud patterns.
2. Calculate fraud-related KPIs.
3. Evaluate authentication mechanisms such as CVV, AVS, and 3D Secure.
4. Identify high-risk users and regions.
5. Visualize fraud trends over time.
6. Generate business recommendations for fraud prevention.

3. Tools and Technologies:

Tool	Purpose
MySQL	Data storage and SQL analysis
SQL Views	Aggregated fraud metrics
Power BI	Dashboard development
DAX	Calculated measures and KPIs
Microsoft Bing Maps	Geographic fraud visualization

4. Dataset Overview:

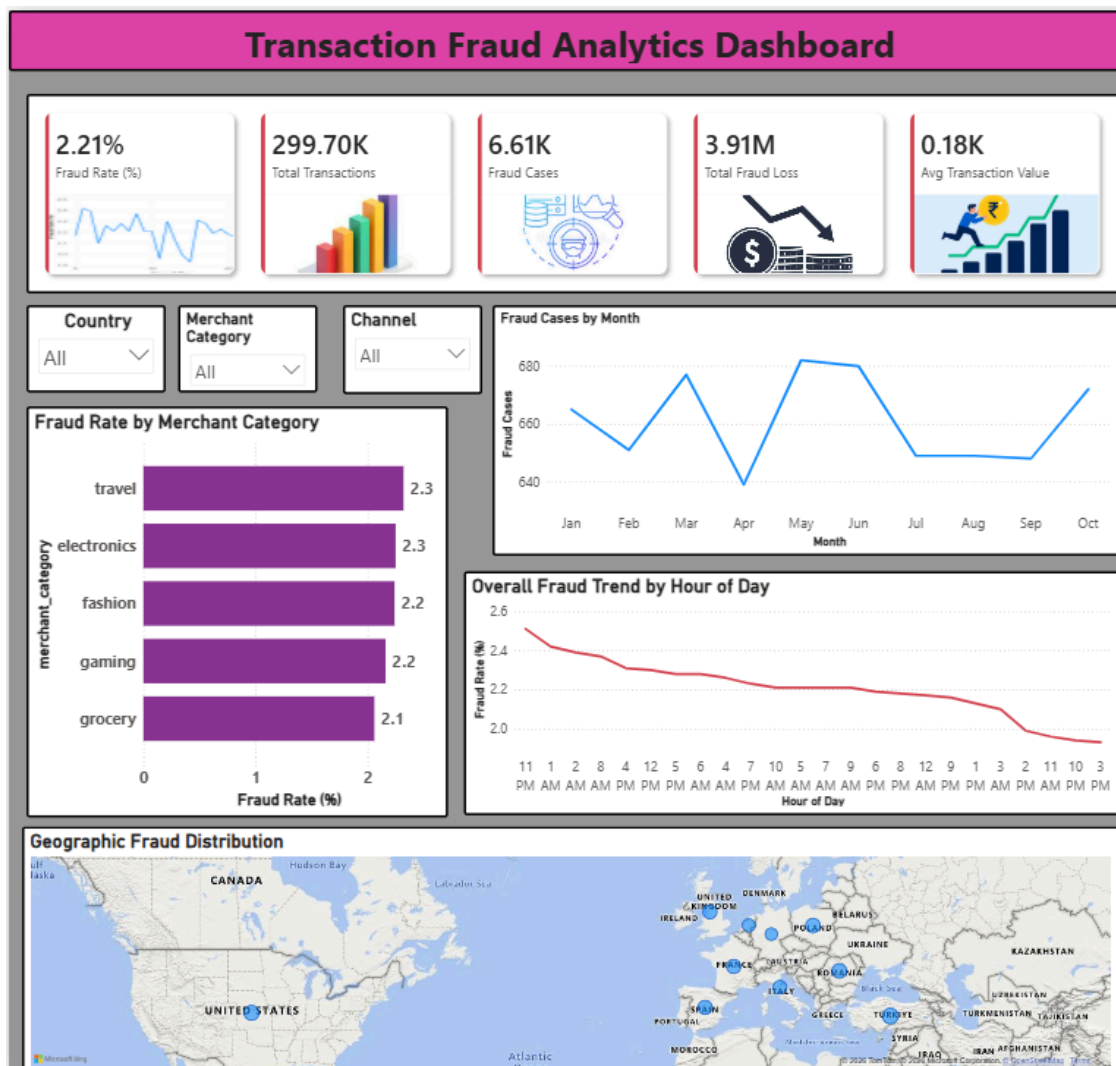
The dataset contains transaction-level information including:

- Transaction Amount
- Merchant Category
- Country
- Payment Channel
- CVV Verification Status
- AVS Verification Status
- 3D Secure Authentication Status
- Fraud Indicator

Several SQL views were created to simplify analytical reporting, including:

- fraud_summary
- user_risk_profile
- merchant_risk
- country_risk
- channel_risk
- monthly_fraud_analysis
- hourly_fraud_analysis
- avs_analysis
- cvv_analysis
- Three_ds_analysis

5. Dashboard 1 – Transaction Fraud Analytics Dashboard



5.1 Key Performance Indicators:

The dashboard provides the following KPIs:

<u>KPI</u>	<u>Value</u>
Fraud Rate	2.21%
Total Transactions	299.70K
Fraud Cases	6.61K
Total Fraud Loss	₹3.91M
Average Transaction Value	₹0.18K

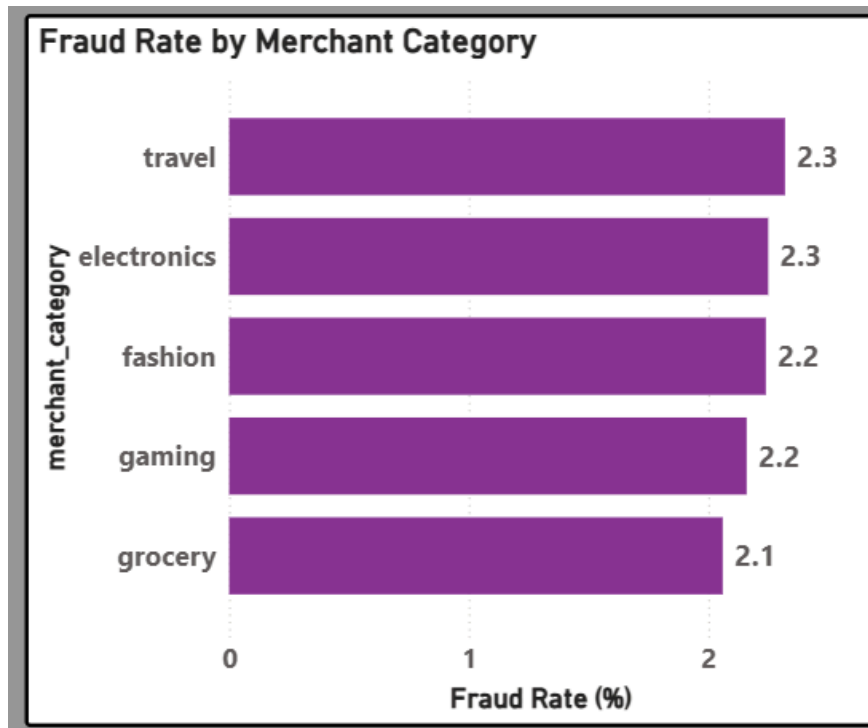
Interpretation:

The overall fraud rate is relatively low at 2.21%; however, the organization has incurred approximately 3.91 million in fraudulent transaction losses, highlighting the financial impact of fraud despite its low occurrence frequency.

5.2 Geographic & Merchant Insights

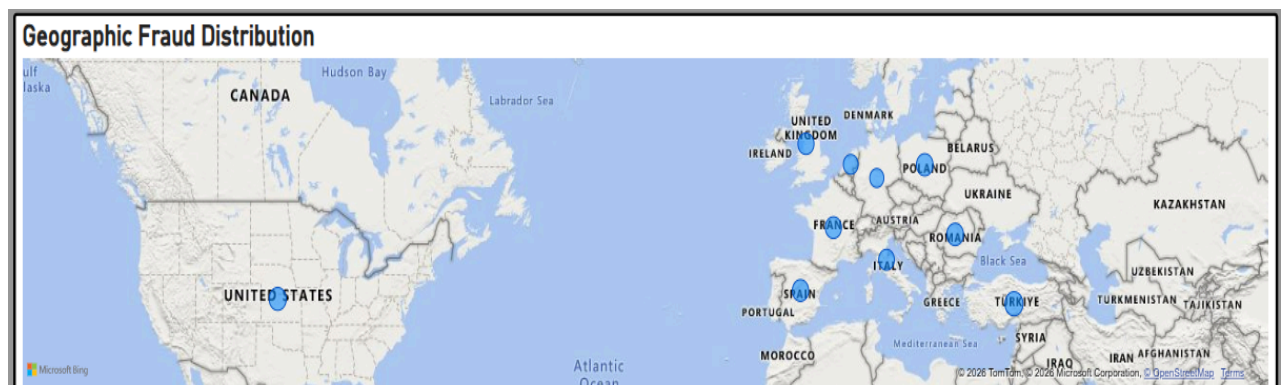
Merchant Analysis:

Travel and Electronics categories exhibit the highest fraud rates among merchant categories analyzed.



Geographic Analysis:

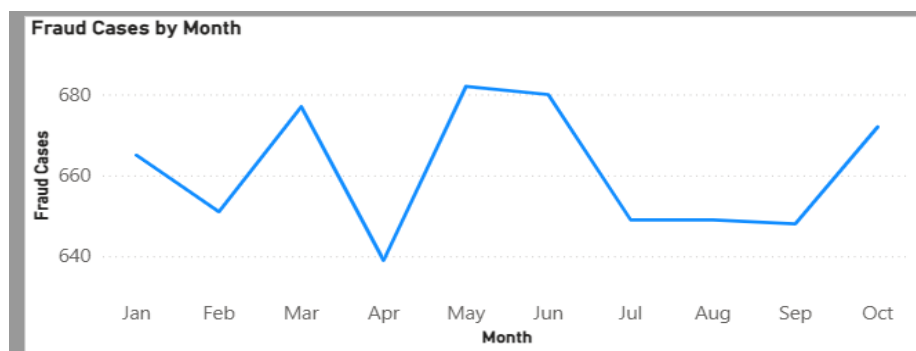
Fraudulent transactions are distributed across multiple countries, indicating the need for geographically aware fraud monitoring strategies.



5.3 Fraud Trend Analysis:

Monthly Trend

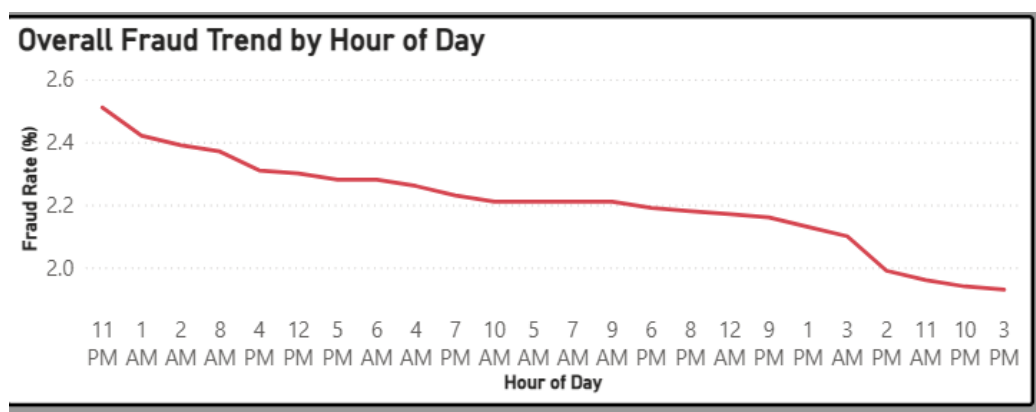
- Fraud cases fluctuate between approximately 640 and 680 cases per month.
- Peak fraud activity occurs during May and June.
- Fraud levels remain relatively stable throughout the year.



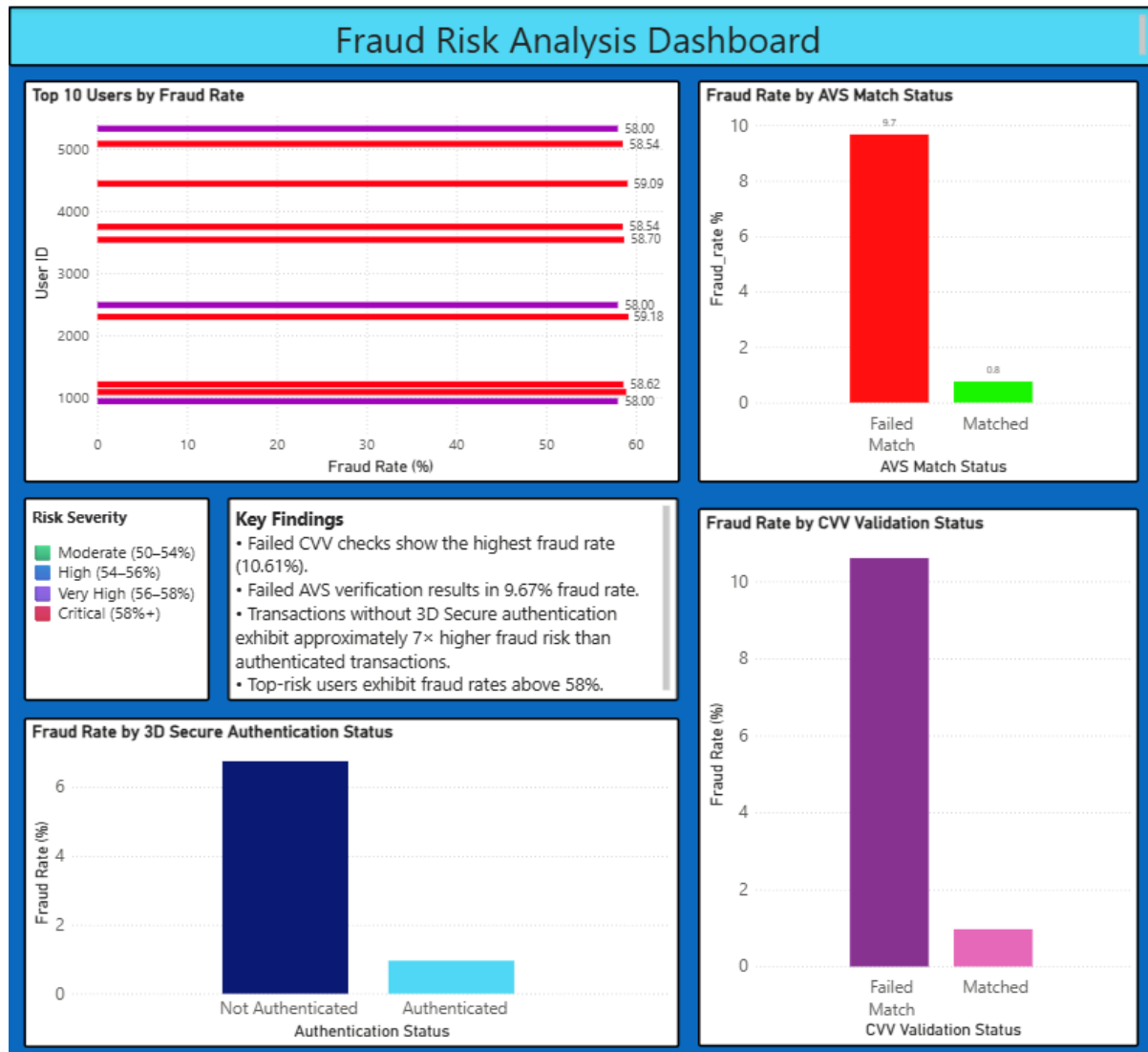
Hourly Trend

Fraud activity varies throughout the day, indicating that transaction timing may contribute to fraud risk patterns.

- Fraud rates vary across different hours of the day.
- Peak fraud activity is observed around late-night hours.
- Lower fraud rates occur during mid-day periods.



6. Dashboard 2 – Fraud Risk Analysis Dashboard:

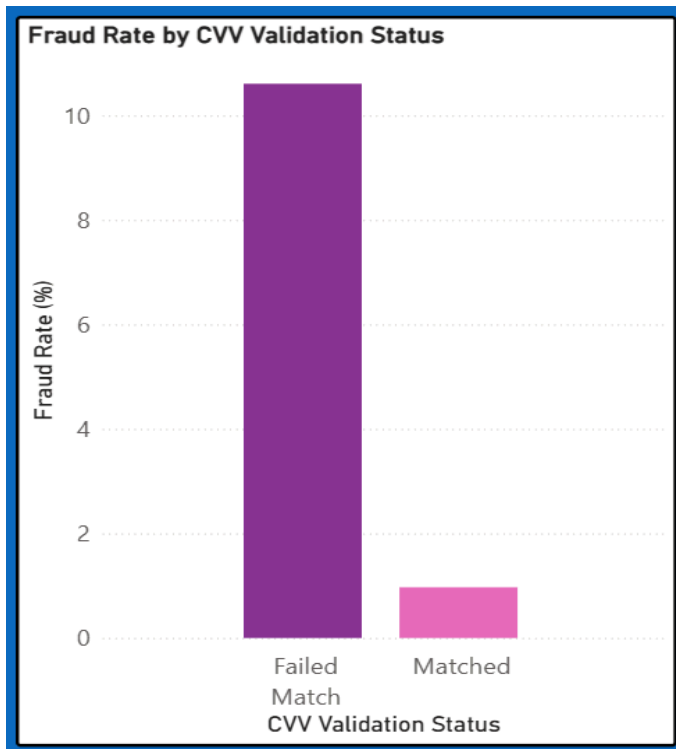


6.1 Failed CVV Validation Is the Strongest Fraud Indicator:

The Fraud Rate by CVV Validation Status chart shows that transactions with failed CVV checks experience significantly higher fraud rates than successfully validated transactions.

This suggests CVV validation is one of the most effective fraud screening mechanisms.

Failed CVV validation exhibits the highest fraud rate at approximately **10.61%**, indicating that card verification failures are a strong fraud indicator.

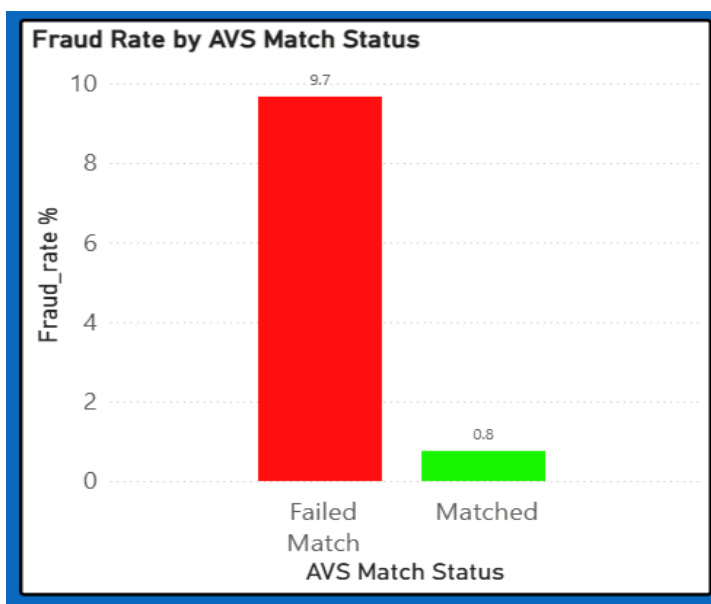


6.2 Failed AVS Verification Significantly Increases Risk:

The Fraud Rate by AVS Match Status chart demonstrates that transactions failing address verification are substantially more likely to be fraudulent.

AVS mismatch should therefore be treated as a high-risk event requiring additional verification.

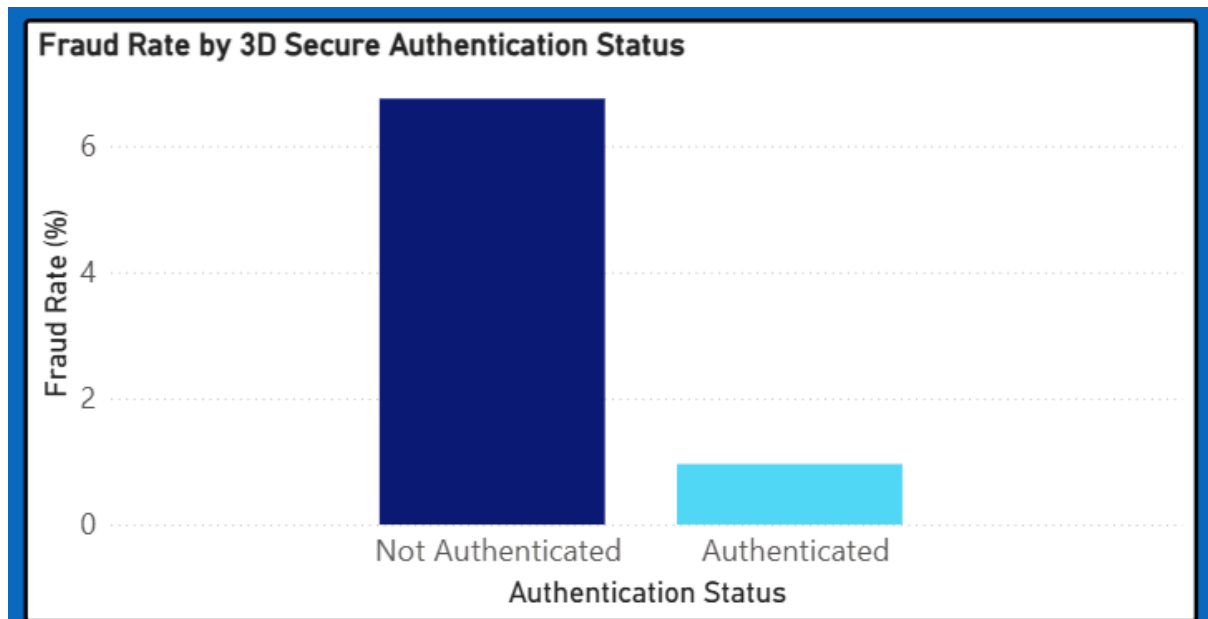
Failed AVS verification also demonstrates elevated fraud risk at approximately **9.67%**.



6.3 3-D Secure Authentication Reduces Fraud Exposure:

The Fraud Rate by 3D Secure Authentication Status chart indicates that authenticated transactions experience considerably lower fraud rates compared to unauthenticated transactions.

This highlights the effectiveness of multi-factor authentication in fraud prevention.



7. Business Recommendations:

1. Strengthen monitoring of failed CVV and AVS transactions.
2. Enforce 3D Secure authentication for high-risk transactions.
3. Introduce additional verification checks for users classified as Critical Risk.
4. Apply enhanced monitoring for Travel and Electronics merchant categories.
5. Implement risk-based transaction scoring for real-time fraud detection.

8. Conclusion:

The Transaction Fraud Analytics Dashboard successfully identifies fraud patterns, quantifies financial impact, and highlights key risk indicators. The dashboard provides decision-makers with a centralized view of fraud activity, enabling proactive fraud prevention and more informed risk management strategies.